

Advanced Linear Algebra: Foundations to Frontiers

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3.1.1 Choosing the right basis ¶

03.1.1 Conditioning of a Vandermonde matrix

fit width



A classic problem in numerical analysis is the approximation of a function, $f: \mathbb{R} \rightarrow \mathbb{R}$, with a polynomial of degree $n - 1$. (The $n - 1$ seems cumbersome. Think of it as a polynomial with n terms.)

$$f(x) \approx \gamma_0 + \gamma_1 x + \cdots + \gamma_{n-1} x^{n-1}.$$

* Now, often we know f only "sampled" at points $x_0, \dots, x_{m-1}$:

$$\begin{array}{rcl} f(x_0) & = & \phi_0 \\ \vdots & & \vdots \\ f(x_{m-1}) & = & \phi_{m-1}. \end{array}$$

In other words, input to the process are the points

$$(x_0, \phi_0), \dots, (x_{m-1}, \phi_{m-1})$$

and we want to determine the polynomial that approximately fits these points. This means that

$$\begin{array}{ccccccc} \gamma_0 & + & \gamma_1 x_0 & + & \cdots & + & \gamma_{n-1} x_0^{n-1} & \approx & \phi_0 \\ \vdots & & \vdots & & \vdots & & \vdots & & \vdots \\ \gamma_0 & + & \gamma_1 x_{m-1} & + & \cdots & + & \gamma_{n-1} x_{m-1}^{n-1} & \approx & \phi_{m-1}. \end{array}$$

⚙ This can be reformulated as the approximate linear system

$$\begin{pmatrix} 1 & \chi_0 & \cdots & \chi_0^{n-1} \\ 1 & \chi_1 & \cdots & \chi_1^{n-1} \\ \vdots & \vdots & & \vdots \\ 1 & \chi_{m-1} & \cdots & \chi_{m-1}^{n-1} \end{pmatrix} \begin{pmatrix} \gamma_0 \\ \gamma_1 \\ \vdots \\ \gamma_{n-1} \end{pmatrix} \approx \begin{pmatrix} \phi_0 \\ \phi_1 \\ \vdots \\ \phi_{m-1} \end{pmatrix}.$$

⚙ which can be solved using the techniques for linear least-squares in **Week 4**. The matrix in the above equation is known as a **Vandermonde matrix**.

⚙ **Homework 3.1.1.1.** Choose $\chi_0, \chi_1, \dots, \chi_{m-1}$ to be equally spaced in the interval $[0, 1]$: for $i = 0, \dots, m-1$, $\chi_i = ih$, where $h = 1/(m-1)$. Write a Matlab code to create the matrix

$$X = \begin{pmatrix} 1 & \chi_0 & \cdots & \chi_0^{n-1} \\ 1 & \chi_1 & \cdots & \chi_1^{n-1} \\ \vdots & \vdots & & \vdots \\ 1 & \chi_{m-1} & \cdots & \chi_{m-1}^{n-1} \end{pmatrix}$$

as a function of n with $m = 5000$. Plot the condition number of X , $\kappa_2(X)$, as a function of n (Matlab's function for computing $\kappa_2(X)$ is `cond(X)`.)

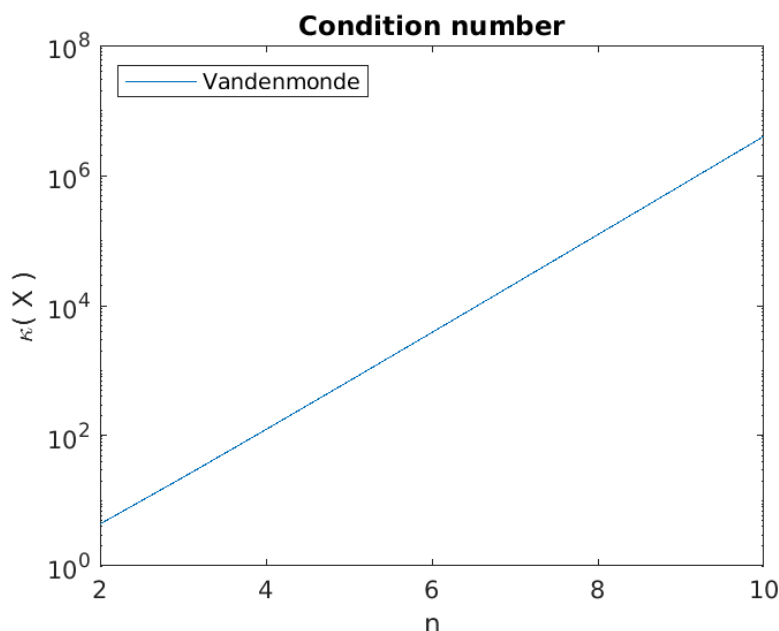
▼ Hint ▼ Solution

- Here is our implementation:

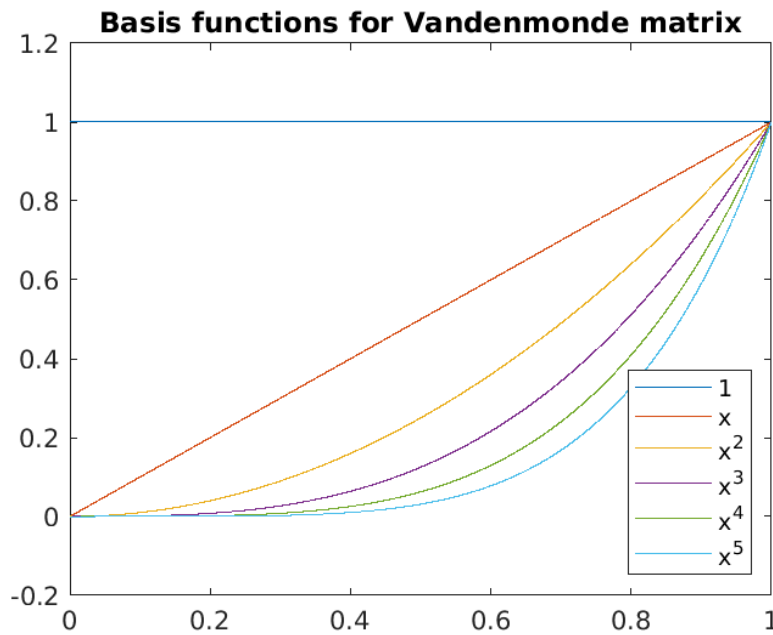
`Assignments/Week03/answers/Vandermonde.m`.

(Assignments/Week03/answers/Vandermonde.m)

- The graph of the condition number, $\kappa(X)$, as a function of n is given by



- The **parent functions** $1, x, x^2, \dots$ on the interval $[0, 1]$ are visualized as



Notice that the curves for x^j and x^{j+1} quickly start to look very similar, which explains why the columns of the Vandermonde matrix quickly become approximately linearly dependent.

Think about how this extends to even more columns of A .

You may want to use the recurrence $x^{j+1} = x x^j$ and the fact that the `.*` operator in Matlab performs an element-wise multiplication.

3 1 1 Legendre polynomials

fit width



An alternative set of polynomials that can be used are known as **Legendre polynomials**. A shifted version (appropriate for the interval $[0, 1]$) can be inductively defined by

$$\begin{aligned}
P_0(\chi) &= 1 \\
P_1(\chi) &= 2\chi - 1 \\
&\vdots \\
P_{n+1}(\chi) &= ((2n+1)(2\chi-1)P_n(\chi) - nP_{n-1}(\chi)) / (n+1).
\end{aligned}$$

☞ The polynomials have the property that

$$\int_0^1 P_s(\chi)P_t(\chi)d\chi = \begin{cases} C_s & \text{if } s = t \text{ for some nonzero constant } C_s \\ 0 & \text{otherwise} \end{cases}$$

☞ which is an orthogonality condition on the polynomials.

☞ The function $f : \mathbb{R} \rightarrow \mathbb{R}$ can now instead be approximated by

$$f(\chi) \approx \gamma_0 P_0(\chi) + \gamma_1 P_1(\chi) + \dots + \gamma_{n-1} P_{n-1}(\chi).$$

☞ and hence given points

$$(\chi_0, \phi_0), \dots, (\chi_{m-1}, \phi_{m-1})$$

☞ we can determine the polynomial from

$$\begin{array}{ccccccc}
\gamma_0 P_0(\chi_0) & + & \gamma_1 P_1(\chi_0) & + & \dots & + & \gamma_{n-1} P_{n-1}(\chi_0) & = & \phi_0 \\
\vdots & & \vdots & & \vdots & & \vdots & & \vdots \\
\gamma_0 P_0(\chi_{m-1}) & + & \gamma_1 P_1(\chi_{m-1}) & + & \dots & + & \gamma_{n-1} P_{n-1}(\chi_{m-1}) & = & \phi_{m-1}.
\end{array}$$

☞ This can be reformulated as the approximate linear system

$$\begin{pmatrix} 1 & P_1(\chi_0) & \dots & P_{n-1}(\chi_0) \\ 1 & P_1(\chi_1) & \dots & P_{n-1}(\chi_1) \\ \vdots & \vdots & & \vdots \\ 1 & P_1(\chi_{m-1}) & \dots & P_{n-1}(\chi_{m-1}) \end{pmatrix} \begin{pmatrix} \gamma_0 \\ \gamma_1 \\ \vdots \\ \gamma_{n-1} \end{pmatrix} \approx \begin{pmatrix} \phi_0 \\ \phi_1 \\ \vdots \\ \phi_{m-1} \end{pmatrix}.$$

☞ which can also be solved using the techniques for linear least-squares in [Week 4](#).

Notice that now the columns of the matrix are (approximately) orthogonal:

Notice that if we "sample" x as $\chi_0, \dots, \chi_{n-1}$, then

$$\int_0^1 P_s(\chi)P_t(\chi)d\chi \approx \sum_{i=0}^{n-1} P_s(\chi_i)P_t(\chi_i),$$

☞ which equals the dot product of the columns indexed with s and t .

☞ **Homework 3.1.1.2.** Choose $\chi_0, \chi_1, \dots, \chi_{m-1}$ to be equally spaced in the interval $[0, 1]$: for $i = 0, \dots, m-1$, $\chi_i = ih$, where $h = 1/(m-1)$. Write a Matlab code to create the matrix

$$X = \begin{pmatrix} 1 & P_1(\chi_0) & \cdots & P_{n-1}(\chi_0) \\ 1 & P_1(\chi_1) & \cdots & P_{n-1}(\chi_1) \\ \vdots & \vdots & & \vdots \\ 1 & P_1(\chi_{m-1}) & \cdots & P_{n-1}(\chi_{m-1}) \end{pmatrix}$$

as a function of n with $m = 5000$. Plot $\kappa_2(X)$ as a function of n . To check whether the columns of X are mutually orthogonal, report $\|X^T X - D\|_2$ where D equals the diagonal of $X^T X$.

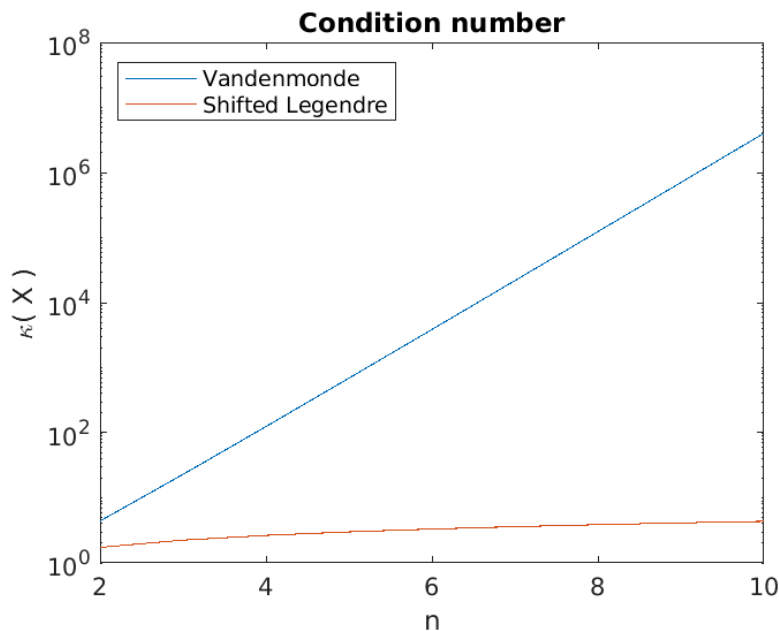
▼ Solution

- Here is our implementation:

[Assignments/Week03/answers/ShiftedLegendre.m](#) .

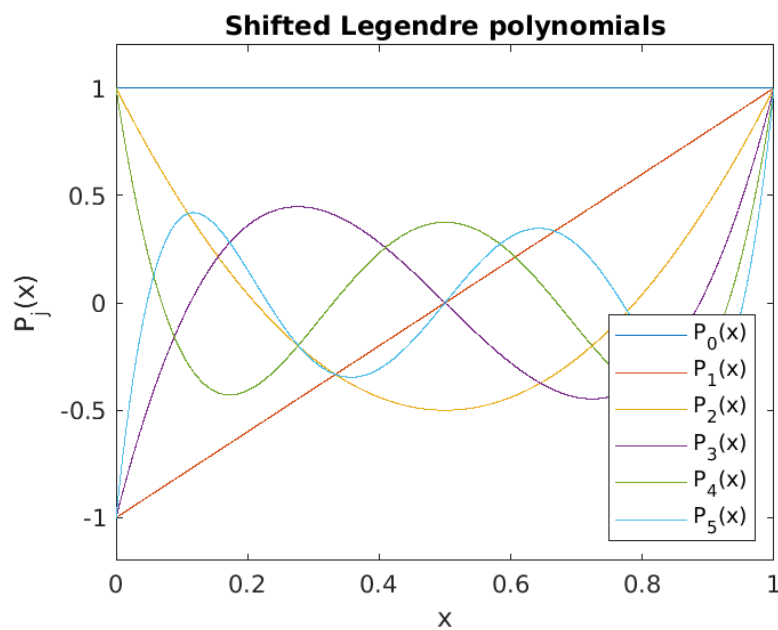
(Assignments/Week03/answers/ShiftedLegendre.m)

- The graph of the condition number, as a function of n is given by



We notice that the matrices created from shifted Legendre polynomials have a very good condition numbers.

- The shifted Legendre polynomials are visualized as



- The columns of the matrix X are now reasonably orthogonal:

$X^T * X$ for $n=5$:

ans =

5000	0	1	0	1
0	1667	0	1	0
1	0	1001	0	1
0	1	0	715	0
1	0	1	0	556

03.1.1 Summary

fit width



Remark 3.1.1.1. The point is that one ideally formulates a problem in a way that already captures orthogonality, so that when the problem is discretized ("sampled"), the matrices that arise will likely inherit that orthogonality, which we will see again and again is a good thing. In this chapter, we discuss how

orthogonality can be exposed if it is not already part of the underlying formulation of the problem.